

Operating Systems Interview Notes

RAG-friendly interview preparation material covering core concepts and explanations.

1. Operating System

An operating system is system software that manages computer hardware and provides services to application programs. It manages processes, memory, files, devices, security, and resource allocation.

2. Types of Operating Systems

Examples include batch systems, time-sharing systems, distributed systems, real-time systems, and network operating systems. Modern operating systems combine several capabilities.

3. Process

A process is a program in execution. It has its own execution state and resources. Process states commonly include new, ready, running, waiting or blocked, and terminated.

4. Process Control Block

A Process Control Block (PCB) stores information needed to manage a process, such as its process ID, state, program counter, CPU registers, scheduling information, and memory information.

5. Threads

A thread is a unit of execution within a process. Threads of the same process can share resources while maintaining their own execution state. Multithreading can improve responsiveness and resource utilization.

6. Process vs Thread

Processes generally have separate address spaces and stronger isolation. Threads within a process share the process address space and resources, making communication easier but requiring careful synchronization.

7. CPU Scheduling

CPU scheduling decides which ready process or thread receives the CPU. Common algorithms include FCFS, SJF, Priority Scheduling, and Round Robin.

8. FCFS and SJF

First Come First Served schedules processes in arrival order. Shortest Job First selects a process with the shortest expected CPU burst and can reduce average waiting time when burst estimates are accurate.

9. Round Robin

Round Robin assigns each ready process a time quantum. After its quantum expires, a running process may move to the end of the ready queue. It is commonly associated with time-sharing systems.

10. Context Switching

A context switch occurs when the operating system saves the state of one running process or thread and loads the state of another. Switching adds overhead because CPU time is spent managing the switch.

11. Synchronization

Synchronization coordinates concurrent processes or threads accessing shared resources. The critical-section problem concerns safe access to shared data.

12. Mutex and Semaphore

A mutex is a mutual-exclusion mechanism that allows controlled access to a shared resource. A semaphore is a synchronization variable that can be used for signaling and controlling access to limited resources.

13. Deadlock

Deadlock occurs when a set of processes waits indefinitely because each process is waiting for a resource held by another. The four classic necessary conditions are mutual exclusion, hold and wait, no preemption, and circular wait.

14. Deadlock Prevention and Avoidance

Prevention attempts to break at least one necessary deadlock condition. Avoidance makes allocation decisions based on whether the system can remain in a safe state. The Banker's algorithm is a classic avoidance approach.

15. Memory Management

Memory management tracks and allocates main memory to processes. Techniques include paging, segmentation, virtual memory, and page replacement.

16. Paging

Paging divides logical memory into fixed-size pages and physical memory into frames. A page table maps pages to frames. Paging avoids the need for a process to occupy contiguous physical memory.

17. Segmentation

Segmentation divides memory according to logical units such as code, data, and stack. Segments can have different sizes and are described by a base and limit.

18. Virtual Memory

Virtual memory allows processes to use an address space larger than available physical memory by keeping parts of the process on secondary storage when necessary.

19. Page Replacement

When a required page is not in memory and no free frame is available, a page replacement algorithm selects a page to remove. FIFO and LRU are common approaches.

20. File Systems

A file system organizes files and directories on storage devices. It manages file names, metadata, allocation, access permissions, and storage space.

21. Disk Scheduling

Disk scheduling determines the order in which disk I/O requests are served. Algorithms include FCFS, SSTF, SCAN, and C-SCAN.

22. Protection and Security

Operating systems use authentication, authorization, permissions, isolation, and other mechanisms to protect resources and data from unauthorized use.